MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)
THIRD SEMESTER PRACTICAL EXAMINATION
(2024 ADMISSION ONWARDS)

MG3DSEBCA200- Basics of Data Analytics using Spreadsheet

Duration of the Examination: **1.5 Hrs** Maximum Marks: **35**

List of Questions

***** All questions belong to the **Apply** level and are aligned with Course Outcome 3 (**CO3**).

1. Open a new Excel workbook. Create a worksheet with sample sales data including: Product, Price, Quantity, and Total Sales.
   - a) Format the headers using bold and background color.
   - b) Use a formula to calculate Total Sales (=Price × Quantity).
2. Use the following data and apply the appropriate function:
   12, 45, 23, 67, 34, 89, 21
   - a) Find the SUM, AVERAGE, MIN, MAX, and ROUND the average to 2 decimal places.
3. Demonstrate the difference between **relative**, **absolute**, and **mixed** referencing with an example (e.g., applying a tax rate to a range of prices).

4. Import a CSV file containing employee records into Excel.
   - a) Remove any duplicate entries.
   - b) Identify and highlight missing data.
   - c) Standardize the date format to DD-MM-YYYY.
5. Use **Text to Columns** to split full names (e.g., “John Smith”) into First Name and Last Name in separate columns.
6. Create a validation rule to restrict user input in the "Age" column to values between 18 and 60.
7. Given a list of email addresses, extract the domain name using **RIGHT**, **MID**, and **FIND** functions.

8. Given a dataset of student scores, calculate:
   - a) Mean, Median, and Mode
   - b) Range, Variance, and Standard Deviation
9. Create a frequency distribution for ages in a dataset and generate a corresponding histogram using Excel’s charting tools.
10. Enable the **Data Analysis Toolpak**. Use it to perform a basic descriptive statistics analysis on a given dataset. Interpret the output.

11. Use an IF function to determine whether a student has "Passed" or "Failed" based on a score threshold of 40.
12. Combine **IF** and **AND** to return "Bonus Eligible" if an employee’s sales are above 50,000 and years of service are more than 5.

13. Use **VLOOKUP** to fetch an employee's department from a second table based on their ID.
14. Using **INDEX** and **MATCH**, retrieve the price of a product from a product list table.
15. Apply **SUMIFS** to calculate total sales for a specific region and month.

16. Using a dataset of monthly sales, create the following charts:

- a) Line chart showing sales over months
- b) Bar chart comparing different regions
- c) Pie chart for product-wise contribution
- d) Scatter plot for advertising spend vs sales

17. Create a **PivotTable** summarizing sales by product and region.

- a) Add slicers for interactive filtering
- b) Generate a Pivot Chart from this data

18. Apply **Conditional Formatting** to highlight:

- a) Top 10% of sales
- b) Sales below target in red
- c) Duplicate entries in a column

19. Create a simple **interactive dashboard** showing KPIs (e.g., Total Sales, Avg Sales per Region) using PivotTables, PivotCharts, slicers, and conditional formatting.